Degraded Visual Environment Sensor (DVES) Program Information
Industry Day Release
SOF/PR & Rotary Program Office
AFLCMC/WIU

1.0 INTRODUCTION: The SOF/PR & Rotary Division (WIU) at Robins AFB, Ga. of the Air Force Life Cycle Management Center (AFLCMC) at Wright-Patterson Air Force Base is officially conducting Industry Day in accordance with Federal Acquisition Regulation (FAR) Part 10 to assess industry's ability to perform A-Kit and B-Kit Engineering & Manufacturing Development (EMD), Air Vehicle (AV) Test and Evaluation (T&E), Low-Rate Initial Production (LRIP), retrofit installation activities of Degraded Visual Environment Sensor (DVES) capabilities on the HH-60W fleet, and associated concurrency modifications to HH-60W training devices. The Government is seeking solutions based on technologies that have been matured to Technology Readiness Level (TRL) 8-9 and Manufacturing Readiness Level (MRL) 8-9 and above.

a) This is not a request for a quote, request for proposal (RFP), or an invitation to bid, nor is this to be construed as a commitment by the Government to issue a solicitation or ultimately award a contract, nor does it restrict the Government to a particular acquisition approach. The Government will not reimburse or pay for any information submitted in response to this Industry Day announcement or any follow-up information requests. Please be advised that all submissions become Government property and will not be returned.
b) AFLCMC is not seeking proposals at this time and will not accept nor consider unsolicited proposals.

1.1. Not responding to this Industry Day does not preclude participation in any future RFP. Small businesses are encouraged to provide responses to this announcement to assist AFLCMC in determining potential levels of competition available in the industry and establish a basis for any subsequent small business subcontract plan goal percentages development. An acquisition strategy is still being determined. Market research results will assist the Air Force in determining whether this requirement will be a full-and-open, a small business set-aside, or a sole-source acquisition.

1.2 In accordance with FAR 15.201(e), responses to this notice are not offers and the U.S. Government cannot accept to form a binding contract. It is the responsibility of the interested parties to monitor the System for Award Management (SAM) database at sam.gov for any additional information pertaining to this announcement.

1.3 Respondents should state whether they are interested in being a prime contractor or a subcontractor for this requirement.

2.0 Objective: This Industry Day will gather information on the current experience and capabilities of prospective industry sources to address HH-60W DVES System A-kit and B-kit EMD, AV T&E, LRIP of approximately five (5) kits, installations on the HH-60W aircraft fleet of approximately 96 aircraft (quantity subject to future National Defense Authorization Act adjustments), and associated HH-60W training system concurrency modifications.

3.0 BACKGROUND: The Rotary Program Office is planning a new contract to integrate Degraded Visual Environment Sensor (DVES) operational capability and safety enhancement upgrades into the HH-60W Jolly Green Giant weapon system. These upgrades must be compatible with the existing HH-60W sensors, displays, navigation, and flight management systems. Crucially, the added system weight must not degrade aircraft performance or mission capabilities. The HH-60W is a multi-engine helicopter primarily designed for worldwide Personnel Recovery (PR) and Combat Search and Rescue (CSAR) missions across all conflict levels and during peacetime. It is rapidly deployable and operates from both established and austere bases, capable of performing missions in all environments, day or night, regardless of weather conditions or threat level, including operations in nuclear, biological, and chemical environments.

4.0 DVES System: The HH-60W performs personnel recovery/combat search and rescue (PR/CSAR) missions, often encountering Degraded Visual Environments (DVEs). These conditions, including smoke, sand, dust, fog, rain, clouds, snow, and smog, significantly impair visibility and create hazards during all phases of flight. Enroute, DVEs can lead to Controlled Flight Into Terrain (CFIT) or collisions with obstacles. During takeoff and landing, rotor wash-induced brownout/whiteout conditions can cause loss of visual reference, increasing the risk of dynamic rollover and hard landings, potentially resulting in aircraft damage or loss and personnel casualties. To mitigate these risks, a system is needed to detect and display uncharted terrain and obstacles in both enroute and terminal areas. This DVES system will enhance crew situational awareness and safety by providing real-time/near real-time obstacle and terrain mapping with integrated displays, along with audible and visual warnings to the aircrew, improving crew coordination during restricted visibility approaches and flight.

5.0 Requirements: The Rotary Program Office is in search of a comprehensive DVES solution that meets the following anticipated requirements. The DVES system shall:

KEY PERFORMANCE PARAMETERS	THRESHHOLD	OBJECTIVE
Detect wires	≥ 0.75 km	>4km
Real-time sensor	Any sensor databases shall continuously attempt to update imagery data from onboard sensors for terrain/obstacles within 1 km of the aircraft.	Any sensor databases shall continuously attempt to update imagery data from onboard sensors for terrain/obstacles within 4 km of the aircraft.

Imagery resolution	The system shall allow aircrew to identify enroute and terminal area hazards with dimensions as small as 12 inches.	The system shall allow aircrew to identify enroute and terminal are hazards with dimensions as small as 1 inch.
Real-time display	Sensor/database imagery shall be available for view simultaneously on all cockpit and cabin MFDs with tactical flight symbology overlaid.	Same as threshold.
ATTRIBUTE		
Mission debriefs	Sensor imagery shall be record-able and available for simultaneous playback with overlaid tactical flight symbology.	Same as threshold.

a) The system shall provide real-time displays with latency less than 100ms on all helmet mounted display (MFD), cockpit and cabin multifunction displays (MFD) in conjunction with tactical flight symbology for enhanced aircraft navigation and pilotage capabilities in both day and night DVE conditions.
b) The system shall be at a Technical Readiness Level (TRL) 8-9 technology,
c) The system shall allow aircrew to identify enroute hazards such as terrain, antennas, wires, and poles. System shall allow aircrew to identify terminal area hazards such as poles, vehicles, fence posts, wires, ditches, berms, holes, and large rocks.
d) The DVES system shall provide symbology and cueing that indicates obstacle/obstruction horizontal/vertical/lateral distances from the aircraft during all critical mission segments.
e) The DVES system shall display flight cues that enable aircrews to maintain >/= 75-meters of separation of the aircraft from obstacles and hazards, with an acceptable margin of error of (+) or (-) 2-meters.
f) The DVE system shall detect and display a spatially accurate depiction of obstacles in the area of interest including objects greater than or equal to 2 feet cubed, poles greater than 30 feet tall and 5 inches in diameter, wires greater than 3/8 inch in diameter, depressions greater than 2' x 2' x 1' (length, width, depth) and variations in landing zone terrain greater than 2' in height from a distance of 50 meters and 50' above ground level and provide pilots visual cueing necessary to safely land and take off from a predetermined landing zone in degraded visual environments.
g) System shall integrate with existing HH-60W sensor, display, navigation, and flight management systems to the maximum extent practical.
h) The system shall not be mounted on the aircraft so as to cause or create damage when the full aircraft weight is on ground.
i) Sensor video with symbology overlay shall be record-able and available for simultaneous playback during aircrew debriefing.

j) The system shall be composed of line replaceable units that are accessible, removable, and replaceable by O-level maintenance.
k) The DVE system shall include a capability to build a geo-referenced mapping of obstacles and hazards from collected data to implement a “see-and-remember technique” so that the crew can maintain awareness of obstacles that do not remain within the sensor Field of View (FOV) for as long as the obstacle or hazard remains operationally relevant.
l) Net system weight of total DVES solution under 150 pounds.
m) The solution should model the US Army solution, where possible, in order to produce efficiencies in research, development, procurement and sustainment.